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# UNDERSTANDING SLEEP AND MENTAL WELLBEING

## Mental Health Educational Content Library | Expanded Chapter 08

Purpose. To help readers understand sleep physiology, recognize common sleep problems, develop supportive routines, and know when professional medical or psychological care is needed.

Audience. General adult readers, older adolescents with appropriate support, caregivers, educators, and community health educators.

Publication status. Evidence-informed educational draft. This handbook is not a diagnostic tool or a substitute for individualized professional care. Clinical review, Rwanda-specific referral checks, accessibility review, and human-reviewed translation are required before public release.

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Important: If you are extremely sleepy, avoid driving or operating machinery. If you have severe difficulty breathing during sleep, sudden loss of consciousness, or other acute symptoms, seek appropriate medical care.

# Learning goals and chapter map

After this chapter, a learner should be able to describe the main stages and regulators of sleep, recognize common patterns of sleep difficulty, explain the relationship between sleep and emotional wellbeing, identify supportive daily habits, and seek appropriate professional assessment when needed.

| Section | What readers will learn |
|---|---|
| 1-4 | Why sleep matters; circadian rhythms; sleep pressure; NREM and REM sleep |
| 5-9 | Sleep needs; inadequate sleep; insomnia and other disorders; mental health connections |
| 10-14 | Practical sleep habits; contextual barriers; treatment, medications and safety |
| 15-19 | Case studies, guided exercises, assessments, answers, glossary and references |

# 1. What is sleep, and why do we need it?

Sleep is a natural, recurring state in which the brain and body undergo coordinated changes. It is not simply a period of inactivity. During sleep, the brain cycles through different stages that support attention, learning, memory, emotional processing and physical functioning. The value of sleep is not measured by duration alone: timing, continuity and how rested a person feels during the day also matter. [1-3]

A single difficult night is common and does not necessarily indicate a disorder. Sleep becomes a greater concern when problems persist, are distressing, or interfere with safe and effective daily life. A person with adequate time in bed may still get poor-quality sleep because of repeated awakenings or an underlying disorder.

## Why the topic belongs in mental health education

Sleep and mental health affect each other. Stress, anxiety, depression, trauma-related symptoms and other difficulties may interfere with sleep. Repeated poor sleep can also worsen concentration, irritability, emotional regulation and coping capacity. This does not mean that every person who sleeps poorly has a mental illness. [3,4]

# 2. The body clock: circadian rhythm

The circadian rhythm is an approximately 24-hour timing system that influences when we feel alert, sleepy, hungry and ready for activity. Light, daily activity and regular waking patterns help synchronize it. The brain receives information about environmental light and uses it to coordinate the sleep-wake cycle. The hormone melatonin typically rises as darkness approaches, helping signal nighttime. [2]

For example, a student who stays awake on a phone until very late and sleeps until noon on weekends may find it difficult to become sleepy at an earlier school-night bedtime. However, teenagers naturally tend toward later sleep timing than younger children; this biological shift should not be mistaken for

laziness.

### Sleep pressure

A second process is the drive to sleep that builds while we are awake. Sleep pressure generally increases as waking hours pass and decreases during sleep. Together, sleep pressure and the circadian rhythm influence when we fall asleep and wake up. Caffeine can temporarily reduce our sense of sleepiness but does not replace the restorative functions of sleep. [2]

## 3. What happens during the night?

Typical sleep contains both non-rapid-eye-movement (NREM) and rapid-eye-movement (REM) stages, cycling several times. Patterns change with age and across the night. [5]

| Stage | Broad description |
|---|---|
| NREM 1 | A light transition from wakefulness into sleep; easy to awaken. |
| NREM 2 | More established sleep; body functions slow and external awareness decreases. |
| NREM 3 | Deep or slow-wave sleep; especially prominent earlier in the night. |
| REM | Brain activity changes; vivid dreams are common and memory/emotional functions are supported. |

No single stage is the only "important" stage. Normal sleep includes a balance across stages, and consumer sleep trackers cannot reliably diagnose a sleep disorder based solely on their stage estimates.

## 4. How much sleep is generally recommended?

Recommended sleep amounts depend on age, and they describe population guidance rather than a rigid score for each individual. Some people need more sleep when ill or recovering from lost sleep. [1,6]

| Age group | Typical recommended sleep |
|---|---|
| 6-12 years | 9-12 hours per 24 hours |
| 13-17 years | 8-10 hours per 24 hours |
| 18-60 years | At least 7 hours, commonly 7-9 hours |
| 61-64 years | 7-9 hours |
| 65 years and over | 7-8 hours |

A person who works long shifts, shares a crowded home, experiences nighttime noise, or cares for a young child may have limited control over sleep time. Educational advice must acknowledge these real

barriers instead of blaming the individual.

# 5. What is insufficient or poor-quality sleep?

Insufficient sleep means obtaining less sleep than the body needs. Sleep deficiency can also involve poor timing, interrupted sleep, or a condition that reduces restorative sleep. Possible daytime consequences include low alertness, difficulty remembering information, slower reaction time, changes in mood and greater risk during driving or hazardous tasks. [3]

## Common signs worth noticing

- Frequently feeling sleepy or unintentionally dozing during the day.
- Struggling to concentrate at school, work or during conversations.
- Needing much more time than usual to complete simple tasks.
- Becoming more irritable or emotionally reactive after poor nights.
- Feeling unrefreshed despite giving yourself adequate opportunity to sleep.

Symptoms can also arise from anemia, medication side effects, other illnesses, depression or other conditions. A professional assessment can help determine the cause.

# 6. Insomnia: more than one bad night

Insomnia involves difficulty falling asleep, staying asleep or returning to sleep after waking, despite an adequate opportunity to sleep, with associated distress or daytime impairment. Temporary insomnia may follow stress, illness, travel, grief or major changes. Chronic insomnia is a more persistent clinical pattern and should not be diagnosed from a simple website checklist. [7-9]

## Common maintaining patterns

A person who worries about whether they will sleep may become increasingly alert at bedtime. Spending extra hours awake in bed, monitoring the clock, or irregular sleep timing can sometimes maintain the problem. Persistent insomnia may also coexist with pain, anxiety, depression, substance use or other sleep disorders.

> Example: A university student spends several nights awake before examinations. After the examination ends, the student keeps worrying about sleep and begins staying in bed much longer to "catch up." The difficulty continues. A clinician can assess whether insomnia-focused treatment is appropriate.

# 7. Other sleep conditions people should know about

## Obstructive sleep apnea

Sleep apnea may involve repeated breathing interruptions, loud snoring, gasping or choking during sleep, and daytime sleepiness. Not everyone with sleep apnea snores. People with these symptoms should seek medical assessment; sleep hygiene alone cannot treat the underlying breathing disorder.

### Circadian rhythm sleep-wake difficulties

Some people cannot align their natural sleep timing with work, school or social schedules. Shift work, jet lag and delayed sleep timing may contribute. Treatment depends on the pattern and may require specialist guidance.

### Restless legs, parasomnias and hypersomnolence

Uncomfortable sensations that improve with moving the legs, unusual behaviors during sleep, or severe daytime sleepiness deserve evaluation. Falling asleep unexpectedly or experiencing sleep attacks should not be assumed to result only from a busy schedule.

## 8. How sleep and mental health interact

### Stress and worry

When the body is on alert, it can be harder to settle at night. Sleep loss can make daily stressors feel harder to manage. Gentle routines may help, but people facing ongoing insecurity or hardship also need practical support.

### Anxiety

Worry about future events or sleep itself may prolong wakefulness. Persistent sleep difficulties and anxiety symptoms should be assessed together rather than treated as unrelated issues.

### Depression

Depression may be associated with difficulty sleeping, early waking or sleeping much longer than usual. Changes in sleep alone do not establish depression, and treatment decisions require careful assessment.

### Trauma-related symptoms

Nightmares, hypervigilance or a feeling of unsafety may disrupt rest. Survivors should not be pressured to close their eyes, disclose memories, or perform relaxation exercises that increase distress.

### Bipolar disorder

A markedly reduced need for sleep accompanied by unusually high energy, elevated or irritable mood, racing thoughts or impulsive behavior can be a warning sign that needs prompt professional assessment. This is different from ordinary insomnia accompanied by fatigue.

## 9. Everyday habits that can support sleep

Healthy sleep routines are useful for general wellbeing and can complement clinical care. They are not, on their own, an adequate treatment for chronic insomnia disorder. [1,8-10]

- Choose a wake-up time that is as consistent as your responsibilities allow.
- Seek natural daylight and appropriate activity during the day.

- Build a predictable winding-down routine: reading, quiet conversation, gentle music or another calming activity.
- Reduce bright screen use close to bedtime if feasible, especially when it keeps you alert.
- Limit caffeine in the afternoon and evening if it affects you.
- Keep the sleep space as dark, quiet, comfortable and safe as reasonably possible.
- Avoid using alcohol as a sleep aid: it can fragment sleep and has other risks.
- If an environment is unsafe or crowded, focus on feasible improvements and seek practical support rather than blaming yourself.

# 10. Practical barriers in Rwanda-relevant daily life

A reader may share one room with family members, work evening shifts, study under difficult conditions, manage caregiving duties or have limited privacy. Advice should prioritize realistic actions. For example, a household may coordinate a quieter period, reduce unnecessary night lighting, or plan a more reliable morning routine when possible. These are illustrative situations, not claims about all Rwandan households.

When poverty, housing insecurity, severe stress, violence or illness disrupt sleep, an individual routine cannot resolve the underlying problem. Health education should connect people to medical, social and practical support where appropriate.

# 11. Evidence-based care for persistent insomnia

## What is CBT-I?

Cognitive Behavioral Therapy for Insomnia (CBT-I) is a structured treatment that addresses sleep-related patterns of thinking and behavior. It can include education, tailored behavioral strategies, cognitive approaches and follow-up. Multiple clinical guidelines recommend multicomponent CBT-I as first-line treatment for chronic insomnia in adults. [8,9]

Specialist CBT-I may include methods such as sleep restriction or sleep compression. These should not be copied from a handout without assessment, especially for people with certain medical or psychiatric conditions, excessive daytime sleepiness, high-risk work, or complex sleep problems.

## Medication and supplements

A clinician may consider medication depending on symptoms, age, other conditions, pregnancy status, interactions and treatment history. Sleeping pills can have side effects and may not address the underlying cause. Do not start, combine, stop or change prescribed medicines on the basis of this chapter. Melatonin is not a universal cure for insomnia and suitability varies. [7,10]

# 12. When should someone seek professional help?

Consider an appointment with a qualified healthcare professional when sleep difficulty persists, significantly affects mood, concentration, study or work, or occurs alongside loud snoring, breathing interruptions, repeated nightmares, unusual movements, severe daytime sleepiness, chronic pain or symptoms of a mental health condition.

Urgent medical assessment is warranted if a person has severe or unusual breathing problems, episodes of losing consciousness, serious injury associated with sleepiness, or new concerning symptoms. If a person is dangerously sleepy, they should not drive or operate machinery. If there are thoughts of suicide or immediate danger, seek urgent emergency assistance through available local emergency services or the nearest emergency department.

Where symptoms suggest mania or a significantly reduced need for sleep with marked behavioral changes, seek prompt clinical assessment.

# 13. Four everyday learning scenarios

## Case 1 - Examination week

A secondary-school learner studies very late, uses a bright phone in bed and struggles to wake for class. The learner feels irritable and forgetful.

Reflection: What realistic changes to study timing and evening routine might help? How might teachers or caregivers support the learner?

## Case 2 - Shift work

A healthcare worker rotates between late and early shifts and becomes sleepy during travel. Family responsibilities limit recovery time.

Reflection: Which barriers require schedule or workplace support? Why is drowsy travel a safety concern?

## Case 3 - Persistent worry

An adult worries about falling asleep and checks the clock throughout the night. Sleep problems continue for weeks and affect work.

Reflection: What makes this different from a single sleepless night? Why might CBT-I assessment help?

## Case 4 - Nighttime breathing concerns

A family notices that one adult snores loudly, sometimes gasps during sleep and remains very tired during the day.

Reflection: Why should this person seek medical assessment rather than only trying herbal tea or an earlier bedtime?

# 14. Guided activity A - My seven-day sleep observation

For seven days, readers may record sleep patterns without trying to force a target. This exercise is for awareness, not diagnosis or clinical sleep restriction. Avoid recording sensitive details on devices that are shared or monitored.

| Day | Approx. bedtime / wake time | Restfulness and daytime notes |
|---|---|---|
| Mon | _____ | _____ |
| Tue | _____ | _____ |
| Wed | _____ | _____ |
| Thu | _____ | _____ |
| Fri | _____ | _____ |
| Sat | _____ | _____ |
| Sun | _____ | _____ |

After the week, consider whether patterns change with shift work, phone use, stress, noise, illness, caffeine or caregiving. Bring the notes to a clinician if a sleep problem is ongoing. If recording sleep increases anxiety, stop tracking.

# 15. Guided activity B - A gentle evening routine

Choose one or two realistic steps, not a rigid checklist: (1) identify tomorrow's priority tasks earlier in the evening, (2) choose a quiet activity for the final part of the day, (3) adjust brightness or noise where feasible, and (4) allow yourself to rest without checking the clock repeatedly.

Reflection: Which step is feasible in your home? What barrier would require someone else's cooperation? A routine should be adapted to safety, disability, caregiving and work schedules.

# 16. Guided activity C - Distinguishing a worry from a plan

Write down an ordinary concern earlier in the evening if doing so feels comfortable. Divide it into "something I can act on tomorrow" and "something I cannot solve tonight." Choose one practical next action for tomorrow. Do not use this exercise to dismiss real danger, pain or urgent needs.

> Example: "I may be late for work" can become "I will prepare my bag and check my transport plan." If a concern involves an unsafe environment or a serious medical symptom, practical safety assistance takes priority over relaxation.

## 17. Knowledge assessment

1. Does the need for sleep change across ages?

2. Is sleep quality relevant even when total hours seem adequate?

3. What two major processes influence sleep timing?

4. Does every difficult night indicate insomnia disorder?

5. Why can late caffeine use be relevant?

6. Is sleep hygiene alone the recommended treatment for chronic insomnia?

7. What symptoms suggest possible sleep apnea?

8. Should someone copy sleep restriction instructions from an article?

9. Can emotional distress and poor sleep affect each other?

10. What should a dangerously sleepy driver do?

## 18. Assessment answers and explanations

1. Yes. Children and teens usually need more sleep than adults.

2. Yes. Fragmented or mistimed sleep can still impair functioning.

3. Sleep pressure and circadian rhythm.

4. No. Isolated sleep difficulty is common.

5. It can interfere with the ability to feel sleepy or fall asleep.

6. No. Multicomponent CBT-I is recommended as first-line treatment for many adults.

7. Snoring, observed breathing pauses or gasping, and excessive daytime sleepiness.

8. No. Such treatment should be tailored and supervised.

9. Yes, the relationship may be bidirectional.

10. Avoid driving and seek appropriate help.

## 19. Glossary

Circadian rhythm: An approximately 24-hour cycle regulating alertness and many bodily functions.

Sleep pressure: The biological tendency to become sleepier with time awake.

REM: A sleep phase associated with characteristic eye movements and often vivid dreams.

NREM: Non-rapid-eye-movement sleep, including lighter and deeper stages.

Insomnia: Difficulty sleeping despite an opportunity to sleep, causing distress or impairment.

CBT-I: Cognitive Behavioral Therapy for Insomnia, a structured evidence-based treatment.

Sleep apnea: A sleep-related breathing disorder that requires medical assessment.

Sleep hygiene: Daily routines and environmental practices that can support healthy sleep.

## Scientific and institutional references


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## Pre-publication review checklist

- Clinical review by appropriately qualified mental health or sleep-care professional.
- Confirm that guidance for children, pregnancy, older adults and shift workers is suitable.
- Rwanda-specific verification of available health services and emergency pathways.
- Kinyarwanda and French translation by qualified reviewers, with terminology validation.
- Accessibility check for headings, contrast, screen readers and printable activities.

- Editorial fact-check of references, updates and any reuse permissions.

This document is educational and does not provide individualized diagnosis, medication instructions, or an emergency service.